Forensic Process Documentation and Reporting

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Learning objectives

- Evaluate the best practices for writing a digital forensics report
- Critically discuss the decision making in relation to the report
- Critically discuss how to apply the scientific method to the report

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Process considerations

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The process

ISO/IEC 27037:2012Guidelines for identification, collection, acquisition and preservation of digital evidence

1. **Identification**
   i. Recognise potential digital evidence
   ii. Determine relevance and scope
2. **Collection**
   i. Gather physical and logical evidence sources
   ii. Secure items appropriately
3. **Acquisition**
   i. Create forensic copies
   ii. Use validated tools
   iii. Verify with hashing
4. **Preservation**
   i. Maintain integrity
   ii. Protect against alteration
   iii. Maintain chain of custody

Practice – oriented stages

ISO 27041 :2015 - Guidance on assuring suitability and adequacy of incident investigative method

ISO 27042: 2015 - Guidelines for the analysis and interpretation of digital evidence

ISO 27043: :2015 - Incident investigation principles and processes

5. **Examination**
   i. Extract relevant artefacts from acquired evidence
   ii. Recover files, metadata, logs, Registry data, memory artefacts
   iii. Validate integrity and maintain chain of custody
   iv. Ensure repeatability of technical steps
6. **Analysis**
   i. Correlate artefacts across multiple sources
   ii. Construct timelines of events
   iii. Identify anomalies and patterns
   iv. Test hypotheses against evidence
7. **Interpretation**
   i. Contextualise findings within the investigative scope
   ii. Distinguish fact from inference
   iii. Consider alternative explanations
   iv. Evaluate reliability and limitations
8. **Reporting**
   i. Present findings clearly and objectively
   ii. Separate evidence from opinion
   iii. Document methodology and tools used
   iv. State assumptions, uncertainties, and constraints

NIST Computer Forensics Tool Testing (CFTT) program

ISO 27037:2012 - Guidelines for identification, collection, acquisition and preservation of digital evidence

Tools

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3 generations of tools

1. Command line tools for examining disk images
   - e.g. The Sleuth Kit, image verification

2. GUI and combined functionality tools
   - e.g. Autopsy, FTK, EnCase
   - imaging, acquisition, analysis, reporting

3. Remote access and live investigations

Types of computer forensics tools

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Hardware or Software

- Evidence acquisition
  - Imaging devices and applications
- Evidence retention
  - Secure cabinets and filing/document management systems
- Secure evidence disposal
  - Shredders and secure deletion software

Commercial or Open Source

Raw data handling applications

- Data 'downloaders'/ importers/ 'formaters'
  - Conversion of large text-based files (e.g. database/spreadsheet)
  - Display of proprietary file formats (e.g .dbx, .pst etc.)
- Hex- and other powerful editors
  - Binaries and mixed-type files investigation
  - Registry information decoding

Types of computer forensics tools (2)

- GUI Forensic Tools
  - Advantages
    - Can simplify forensic investigations
    - Easy to learn
    - Multi-tasking
    - Contain a suite of tools
  - Disadvantages
    - Excessive resource requirements i.e. large amounts of processing power
    - May produce inconsistent results because of OS used

Good Sources:

- Mares and Company
- Eric Zimmerman's tools

How to choose a digital forensics tool?

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1. What OS does the forensic tool work on?

2. Can it analyse more than one file system?

3. Can a scripting language be used with the tool to automate repetitive functions & tasks?

4. Does the forensic tool have any automated features that can help reduce the data analysis time?

5. Does the vendor provide any product support?

Acquisition tools

- Physical copy of entire disk drive
- Logical copy of disk partition
- Bit stream image
- Segmenting feature as easier to store data on smaller media
- Vendor Specific data formats that compress data to reduce storage size - typically 50% less
- Verification methods using hash values of acquired
- Remote, live, and memory acquisitions

Validation, verification & discrimination

- Validation
  - Process ensures integrity of data and functionality of tool
- Verification
  - Obtain hash value of entire disk or hash values of individual files
  - Produces a unique hexadecimal value similar to a fingerprint
  - Used to make sure original data hasn't changed
- Discrimination of data involves sorting & searching through data
  - Purpose to remove good data from suspicious data
  - Good data is OS files, common programs, etc.
  - Good data file hashes integrated into software
  - Used to eliminate good data automatically
  - Analysing file headers of known file types

Extraction

- Recovery of data is first step in data analysis
- Data viewing mechanism for digital evidence
- Keyword search
- Analyse unallocated data area of disk drive
  - Data Carving
- Bookmarking data for future reference & reporting
- Decrypting files or systems

Reconstruction

- Disk to disk copy adjusts target disk drive geometry to match the original drives, cylinders, etc
  - Hardware duplicators
  - Disk-to-disk copy
  - Partition-to-partition copy
  - Image-to-disk copy
  - Image-to-partition copy
  - Disk-to-image copy
  - Rebuilding files from data runs and carving

Reporting

- Most digital forensics tools have a reporting function
- Record of activity performed by forensic examiner – log report
  - Bookmarking or tagging
  - Log reports
  - Timelines
  - Report generator

General considerations

- Technology changes rapidly
- Hardware eventually fails
  - Schedule equipment replacements periodically
- Carefully consider what you need
- Consider the type of your investigations
- Consider your systems
- Test your tools and the always verify the results they produce

Lab considerations

- Your lab must meet the following criteria
  - Establish categories for digital forensics tools
  - Identify forensics category requirements
  - Develop test assertions
  - Identify test cases
  - Establish a test method
  - Report test results

Lab Requirements

- Safe and secure physical environment
- Secure facility
  - Should preserve integrity of evidence data
- Minimum requirements
  - Small room with true floor-to-ceiling walls
  - Door access with a locking mechanism
  - Secure container
  - Visitor's log
- People working together should have same access level
- Brief your staff about security policy
- TEMPEST facilities

Documentation & Reporting

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Purpose of the report

A documented process to ensure a repeatable standard is met by the forensic analyst or the organization they are representing.

- *why* the computer system was reviewed?
- *how* the computer data was reviewed?
- *what* conclusions were arrived at?

Digital forensic reports must be technically accurate and easy to read.
A great investigation can be rendered largely ineffective if the resulting report is poor.
As with everything in the field, you have to have a methodology for writing technical reports.

Report Goals

- ✓ Accurately describe the details of an incident
- ✓ Be understandable to decision-makers
- ✓ Be able to withstand a barrage of legal scrutiny
- ✓ Be unambiguous and not open to misinterpretation
- ✓ Be easily referenced
- ✓ Contain all information required to explain your conclusions
- ✓ Offer valid conclusions, opinions, or recommendations when needed
- ✓ Be created in a timely manner

Applying the Scientific Method to Digital Forensics

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Gather information and make observations

Form a hypothesis to explain observations

Examination

Evaluate the hypothesis

Develop possible explanations

Draw conclusions and communicate findings

'if the hypothesis is true, then we would expect to find X in the evidence'

Convey your work to decision makers

The decision makers

- attorneys,
- members of a jury,
- executives in a company

Let the evidence speak for itself - do not leap to conclusions

Testimony

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<u>Expert witness</u>

include their opinions about the evidence found or observed

opinions are based on experience and deductive reasoning extracted from the facts found during an investigation.

<u>Technical witness</u>

only provide the facts found during an investigation

presenting the evidence and how obtained

not allowed to offer opinions or conclusions

General Methodology

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01
Gather the data

02
Analyse the results

03
Outline and Organise the report

04
Write a rough draft

05
Revise the rough draft

Gathering Data

1. Technical report preparation begins with proper planning.
2. Maintain orderly records as the data are gathered.
3. Document investigative steps immediately.
4. Do not use shortcuts or shorthand.
5. Writing clearly and concisely at the moment of evidence discovery promotes accuracy and saves time later:
   - Document as you go!
   - Don't forget – consider how the forensic data should be presented and record the results in this manner.

Analysing the results

- During the analysis and data review, conclusions should be drawn.
  - Limited information gathered during the first phase may lead the forensic analyst to incorrect assumptions.
  - As data is gathered, the conclusions may (and probably will) change.
  - Once the conclusions are drawn, it is best to list them in descending order of importance.

The timeline

- Timeline of key findings to help identify patterns and organize findings
  - provide a useful summary for litigators
  - tabular or visual form
  - link analysis and association matrix to provide decision makers

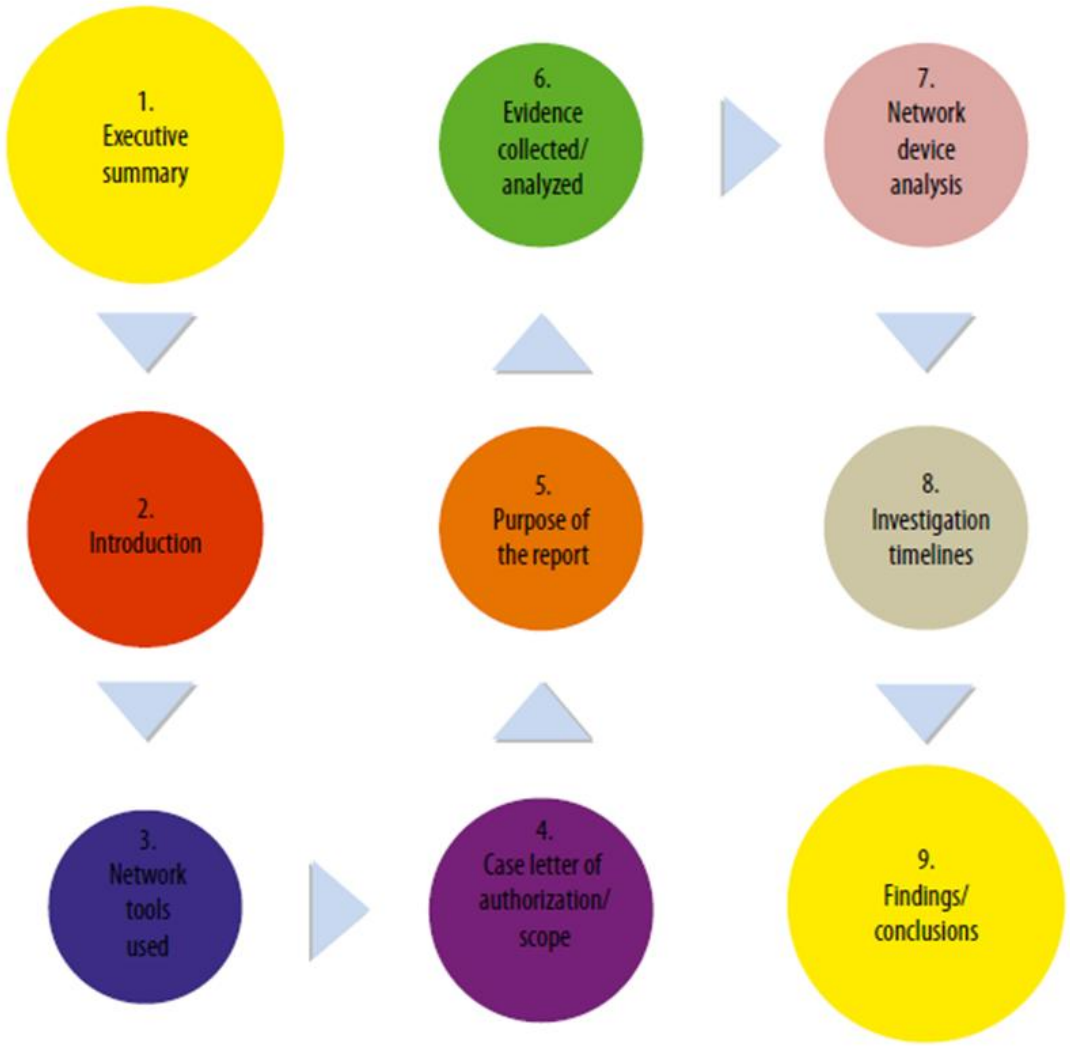
	A	B	C
1	Date/Time	Entry	Source
2			
3	02/16/13 02:34 PM	A USB Device identified as "garmin__nuvi" was successfully installed (2:34~2:40PM)	Setupapi.log & Filesystem
4	02/16/13 02:36 PM	A total of 24 images (DSCN3680.JPG ~ DCN3703.JPG) were stored at My Pictures\Valentines Day folder. Taken with a Nikon Coolpix P4 camera between 2:36:22PM and 2:38:40PM. EXIF header information indicates these pictures were digitized between 6:41PM and 6:56PM on February 14, 2009.	File System & EXIF Header Information
5	02/16/13 02:45 PM	The folder ...\Roman\My Documents\Garmin was created.	File System
6	02/16/13 02:40 PM	Hushmail – Free Email with Privacy - bmoreagent@hushmail.com	Firefox 3 Internet History
7	02/16/13 02:50 PM	Google search for "check ip address". Subsequent access to http://whatismyipaddress.com "Lookup IP, Hide IP, Change IP, Trace IP and more…"	Firefox 3 Internet History
8	02/16/13 03:00 PM	User accessed file C:/Documents and Settings/Roman/My Documents/Garmin/gpx/current.gpx	IE History & "Recent" Links
9			
10	03/20/13 12:32 PM	Google search for "Word Trade Center Baltimore building plans". Subsequent access to http://www.marylandports.com/opsalert/eBroadcast/2008/HPPwtc2008.pdf.	Firefox 3 Internet History
11	03/20/13 01:12 PM	Accessed a Hushmail web page. Welcome to Hushmail Account (bmoreagent@hushmail.me).	File System
12	03/20/13 01:18 PM	Accessed www.gunbroker.com web page "Account is Now Active".	File System & IE History
13	03/20/13 01:18 PM	Accessed http://www.gunbroker.com/Auction/ViewItem.asp?Item=125130891 (SIGARMS, P229, 9MM, NIGHT SIGHTS, 13RD ,2 MAGS).	File System
14	03/20/13 01:19 PM	Accessed http://www.gunbroker.com/Auction/Browse.asp?Cat=3026 (Browsed Pistols->Semi-Auto)	File System
15	03/20/13 01:19 PM	Accessed web page to Ask Seller A Question - Send Mail to User (http://www.gunbroker.com/Auction/SendMailToUserForm.asp?User=398499&Item=1252884862 b67f14c) bmoreagent@hushmail.me, oneclickshooting.com Item 125288486.	File System
16			
17	03/21/13 12:00 PM	Bookmark Created: http://www.google.de/swr?q=undetectable+bomb&hl=de&client=firefox-a&rls=org.mozilla:en-US:official&hs=WiN&swrnum=53600	Firefox 3 Internet History
18			
19	05/24/13 02:17 PM	User engaged in Skype chat with user js-526177 (John Smith) between 2:17 and 2:26PM.	File System and main.db
20	05/24/13 02:21 PM	User again accessed file C:/Documents and Settings/Roman/My Documents/Garmin/gpx/current.gpx.	IE History & File System
21	05/24/13 02:22 PM	User accessed http://login.live.com, Sign In page	IE History
22	05/24/13 02:22 PM	User executed an initial failed Google Search ("we could not understand") on lat ="39.286130" lon ="-76.609936"	File System & IE History

Outlining

- Without the outline, most inexperienced forensic analyst write reports that are confusing and difficult to follow.
- In the analysis stage, concentration was on *what* results should be presented in the forensic report.
- In the outlining stage, concentration is directed on *how* the results should be presented.
- Organising the report is also critically important:
  - start at the high level, and have the complexity of the forensic report increase.
  - This way, the high-level executives need to read only the first page to get a summary of the conclusions.
    - They usually are not interested in the low-level details that support the conclusion.

Outlining: Template

- Executive Summary/ Abstract
- Table of contents
- Objectives
- Body of report
  - Digital Evidence Analysed
  - Relevant Findings
  - Supporting Details
  - Investigative Leads
- Conclusion/ Recommendations
- References
- Glossary
- Acknowledgments
- Appendixes



Writing the draft

When the outline template is created, the first draft should be a straightforward task.

Do not expect to write the report on one try.

The report will be read by technical and non-technical individuals.

Reviewing the draft

1. The material in the report:
   - Are the conclusions valid?
   - Is sufficient information given to support the conclusions?
   - Is enough information given to explain the results?
   - Have all irrelevant ideas been deleted?
   - Are the illustrations necessary?
2. The organization of the report:
   - Are the subject and purpose clearly stated?
   - Does the report flow smoothly from beginning to end?
   - Are the relations between topics clear?
   - Is each illustration clear and properly labelled?
   - Are all required parts of the report included?
3. Spelling and grammar

BS10008:2014

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- British Standards Institution: "Evidential Weight and Legal Admissibility of Electronic Information"
- Emphasises policies for the storage, transfer, and risk assessment of electronic information with a view to its usefulness as evidence in U.K. courts.

the evidential weight of electronic information managed by the corporate body will be maximised, by ensuring its trustworthiness and reliability

Discussion Point

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Example of evidence misinterpretation

A tool attempts to reconstruct remnants of web-based e-mail messages from unallocated space.

This process can result in erroneous combinations of data fragments, giving the impression that someone sent a message that never existed.

1. How would you verify this process?
2. Why do we need a scientific method?

Summary

- Writing a digital forensic investigation report is not an easy task
- Digital forensic reports will be presented in technical and non-technical individuals
  1. They have to be factual
  2. They have to be simple
  3. They have to be concise

Questions?

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Lillard V.T., 2010, Digital Forensics for Network, Internet, and Cloud Computing: A Forensic Evidence Guide for Moving Targets and Data, Elsevier Inc., ISBN 978-1-59749-537-0

Prosise C. and Mandia K., 2003, Incident Response and Computer Forensics, The McGraw-Hill Companies, ISBN: 0-07-222696-X

Appendix

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Not to forget!

Chain of custody and contemporaneous notes!

A recording of every action performed from the time the security incident was detected to its final resolution.

The dating and signing of every document generated and piece of relevant evidence obtained.

The labelling and tagging of evidence gathered during the investigation.

A listing of comments, errors, and omissions as deemed necessary during the evidence collection and analysis process.

Analysing:

- Referencing
  - Developing a consistent way of referencing evidence throughout the report is critical.
  - A good suggestion is to create a unique identifier or reference tag for each person, place, and thing referred to in the forensic report.
- Figures
  - Figures and tables organization should be carefully considered
  - Illustrations are one of the best ways of emphasizing and supporting conclusions.
- Appendices
  - It is important not to interrupt the forensic report with pages and pages of source code right in the middle of a conclusion.
  - Every file that contributes to the conclusion should be included as an appendix to the forensic report.
- Integrity
  - Create and record the MD5 hashes of the evidence as well as record and include the metadata for every file cited in the forensic report.
  - Record and include the metadata for every file or file fragment cited in your report.

Good practise

- Proper planning
- Maintain orderly records
- Document investigative steps
- Document as you go
- Consider the forensic data presentation and record the results in this manner